

Lecture 3 — No-Mercy Practice

10 Python Practice Questions

Q1 — HARD

Take two numbers and print whether they are:

- Both positive
- Both negative
- One positive and one negative
- At least one is zero

Also print which number is greater.

Q2 — HARD

Take a person's age and their exam percentage.

Print:

- "Invalid" if either value is negative.
- "Eligible" if age is at least 18 and percentage is at least 60.
- "Age Requirement Failed" if only the age condition fails.
- "Percentage Requirement Failed" if only the percentage condition fails.
- "Both Requirements Failed" if both fail.

Q3 — HARD

Take three integers and determine whether the middle value is:

- Greater than both others
- Smaller than both others
- Between the other two
- Equal to at least one other value

Handle duplicate values correctly.

Q4 — VERY HARD

Take a four-digit number.

Determine whether:

- The number is actually four digits.
- The first digit is even.
- The last digit is odd.
- The sum of the first and last digits is greater than the sum of the middle two digits.

Print "PASS" only if all conditions are satisfied.

Q5 — VERY HARD

Take a student's marks in three subjects.

Calculate the percentage and classify them:

- 90+ → "A"
- 75–89 → "B"
- 60–74 → "C"
- 40–59 → "D"
- Below 40 → "F"

However, any individual subject below 33 automatically results in "F", regardless of the percentage.

Q6 — VERY HARD

Take three sides of a triangle.

Determine:

1. Whether the sides can form a triangle.
2. If valid, whether it is Equilateral, Isosceles, or Scalene.
3. Whether it is also a right-angled triangle.

Handle invalid and zero/negative sides.

Q7 — VERY HARD

Take a year as input.

Determine whether it is a leap year using the proper rules:

- Divisible by 400 → leap year
- Divisible by 100 → not a leap year
- Divisible by 4 → leap year
- Otherwise → not a leap year

Print "Leap Year" or "Not a Leap Year".

Q8 — NO MERCY

Take a person's age, monthly income, and credit score.

Determine their loan category:

- "Rejected" if age is below 18.
- "Rejected" if income is negative.
- "Premium" if age ≥ 25 , income $\geq 50,000$ and credit score ≥ 750 .
- "Standard" if age ≥ 21 , income $\geq 25,000$ and credit score ≥ 650 .
- "Basic" if age ≥ 18 , income $\geq 15,000$ and credit score ≥ 550 .
- Otherwise "Rejected".

Make sure overlapping conditions produce the correct highest category.

Q9 — NO MERCY

Take three numbers representing a date: day, month, and year.

Determine whether the date is valid.

You must account for:

- Months with 31 days
- Months with 30 days
- February
- Leap years
- Invalid months
- Invalid days
- Negative/zero values

Print only "Valid Date" or "Invalid Date".

Q10 — FINAL BOSS

Build a three-number mathematical analyzer.

Input: number1, number2, number3.

Determine:

- Largest number
- Smallest number
- Whether all three are equal
- Whether exactly two are equal
- Whether all three are different
- Whether the largest is even or odd
- Whether the smallest is positive, negative, or zero
- Whether the sum of all three is positive, negative, or zero

Restriction: Do this using only concepts you have already studied. No lists, loops, functions, or unlearned shortcuts.

Questions: 10 **Solutions:** Hidden **Hints:** Hidden

New concepts intentionally introduced: 0

Difficulty: ■■■■■■